

Graph Database Benchmark System - Backend API Documentation

Base URL

`http://localhost:8080`

1. Select Database

Method

POST

API

`/database/select/COGNODB`

Full URL

`http://localhost:8080/database/select/COGNODB`

Expected Output

Database selected : COGNODB

2. Database Connection

Method

GET

API

`/benchmark/connect`

Full URL

`http://localhost:8080/benchmark/connect`

Expected Output

```
{
  "connected": true,
  "databaseName": "CognODB",
  "message": "Connected Successfully"
}
```

3. Import Dataset

Method

POST

API

/datasets/import

Full URL

http://localhost:8080/datasets/import

Body

form-data

Key	Type	Value
file	File	pokec_edges.csv

Expected Output

```
=====
Dataset Import Completed
=====
File           : pokec_edges.csv
Relationships   : 100000
Time (ms)      : 4528
Time (sec)     : 4.53
=====
```

4. Graph Verification

Method

GET

API

/verify

Full URL

http://localhost:8080/verify

Expected Output

```
=====
Graph Verification
=====
```

Nodes : 44851
Relationships : 84005
=====

5. Traversal Benchmark (1-Hop)

Method

GET

API

/benchmark/traversal/1

Full URL

http://localhost:8080/benchmark/traversal/1

Expected Output

```
{
  "message": "Traversal Benchmark Completed",
  "metric": {
    "benchmarkName": "1-Hop Traversal",
    "averageLatency": 1197.95,
    "minLatency": 1082.05,
    "maxLatency": 1503.04,
    "p50Latency": 1117.63,
    "p95Latency": 1503.04,
    "throughput": 0.834
  }
}
```

6. Traversal Benchmark (2-Hop)

Method

GET

API

/benchmark/traversal/2

Expected Output

```
{
  "message": "Traversal Benchmark Completed",
  "metric": {
    "benchmarkName": "2-Hop Traversal",
    "averageLatency": 1180.73,
    "minLatency": 1131.61,
    "maxLatency": 1215.34,
    "p50Latency": 1194.27,
    "p95Latency": 1215.34,
  }
}
```

```
    "throughput":0.846
  }
}
```

7. Traversal Benchmark (3-Hop)

Method

GET

API

/benchmark/traversal/3

Expected Output

```
{
  "message":"Traversal Benchmark Completed",
  "metric":{
    "benchmarkName":"3-Hop Traversal",
    "averageLatency":1198.95,
    "minLatency":1128.05,
    "maxLatency":1304.70,
    "p50Latency":1199.87,
    "p95Latency":1304.70,
    "throughput":0.834
  }
}
```

8. Point Lookup Benchmark

Method

GET

API

/benchmark/lookup

Expected Output

```
{
  "message":"Lookup Benchmark Completed",
  "metric":{
    "benchmarkName":"Point Lookup",
    "averageLatency":1113.97,
    "minLatency":1097.74,
    "maxLatency":1142.10,
    "p50Latency":1110.60,
    "p95Latency":1142.10,
    "throughput":0.897
  }
}
```

9. Aggregation Benchmark

Method

GET

API

/benchmark/aggregation

Expected Output

```
{
  "message": "Aggregation Benchmark Completed",
  "metric": {
    "benchmarkName": "Aggregation",
    "averageLatency": 508.75,
    "minLatency": 480.06,
    "maxLatency": 578.38,
    "p50Latency": 481.22,
    "p95Latency": 578.38,
    "throughput": 1.965
  }
}
```

10. Mixed Workload Benchmark

Method

GET

API

/benchmark/mixed

Expected Output

```
{
  "message": "Mixed Benchmark Completed",
  "metric": {
    "benchmarkName": "Mixed Workload",
    "averageLatency": 1326.34,
    "minLatency": 479.09,
    "maxLatency": 2729.20,
    "p50Latency": 1106.49,
    "p95Latency": 2729.20,
    "throughput": 0.753
  }
}
```

11. Complete Benchmark Suite

Method

GET

API

/benchmark/run

Expected Output

```
{
  "results": [
    {
      "benchmarkName": "1-Hop Traversal",
      "averageLatency": 1197.16
    },
    {
      "benchmarkName": "2-Hop Traversal",
      "averageLatency": 1129.15
    },
    {
      "benchmarkName": "3-Hop Traversal",
      "averageLatency": 1109.39
    },
    {
      "benchmarkName": "Point Lookup",
      "averageLatency": 1111.52
    },
    {
      "benchmarkName": "Aggregation",
      "averageLatency": 457.60
    },
    {
      "benchmarkName": "Mixed Workload",
      "averageLatency": 846.42
    }
  ]
}
```

12. Benchmark Report

Method

GET

API

/report

Expected Output

```
{
  "databaseName": "CognoDB",
  "generatedTime": "2026-08-06T18:04:14",
  "benchmarks": [
    {
      "benchmarkName": "1-Hop Traversal",
      "averageLatency": 1335.69
    }
  ]
}
```

```
}
```

13. JSON Report

Method

GET

API

/report/json

Expected Output

```
{
  "databaseName": "CognoDB",
  "generatedTime": [2026, 8, 6, 18, 13, 24],
  "benchmarks": [
    {
      "benchmarkName": "1-Hop Traversal"
    }
  ]
}
```

14. CSV Report

Method

GET

API

/report/csv

Expected Output

```
Benchmark,Average,Minimum,Maximum,P50,P95,Throughput
1-Hop Traversal,1167.81,1066.53,1273.40,1142.28,1273.40,0.856
2-Hop Traversal,1088.53,1060.50,1135.50,1086.01,1135.50,0.918
3-Hop Traversal,1260.56,1156.71,1383.07,1207.47,1383.07,0.793
Point Lookup,1235.27,1185.85,1317.54,1196.38,1317.54,0.809
Aggregation,501.68,489.71,514.17,503.74,514.17,1.993
Mixed Workload,1182.20,1137.83,1221.90,1180.69,1221.90,0.845
```

15. Export JSON Report

Method

GET

API

/report/export/json

Expected Output

Saved : reports/benchmark-report.json

16. Export CSV Report

Method

GET

API

/report/export/csv

Expected Output

Saved : reports/benchmark-report.csv

Backend API Summary

Module	APIs
Database Selection	1
Connection	1
Dataset Import	1
Graph Verification	1
Benchmark	6
Report	5
Total APIs	16

This layout is suitable for pasting directly into Microsoft Word as a project/API documentation section.